



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID

Bridge Program

Physics

Lesson 5:

Measurements and units

Class Outline

Title of Unit	Mechanical energy
Title of Lesson	Measurements and units
Starter Activity	Interpret the chart – US gasoline price
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- question: interpret the meaning of the US gasoline price graph - question: various units for length, time and mass - lecture: SI units and 7 SI fundamental units and other derived units - inquiry: scientific notation
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- slow down to explain the video “powers of 10”
	International Mindedness:
	Internationally agreed standard units is a fundamental requirement for scientific collaboration and progress
Lesson Objectives	By the end of the lesson, students will be able to: - know how to derive units from a formula - know the 7 SI base units - write and read numbers in scientific notation
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	- thinking: derive units from formulas
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	International mindedness
Teaching Activities	1. Interpret the chart 2. write a unit for mass time and length 3. Video: powers of 10
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Homework instruction Show steps for unit derivation

Vocabulary



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Unit 单位

Gallon 加仑

Fundamental 基本的

Derive 推导

Scientific notation 科学计数法

Exponent 指数

Expression 表达

Abbreviation 简写

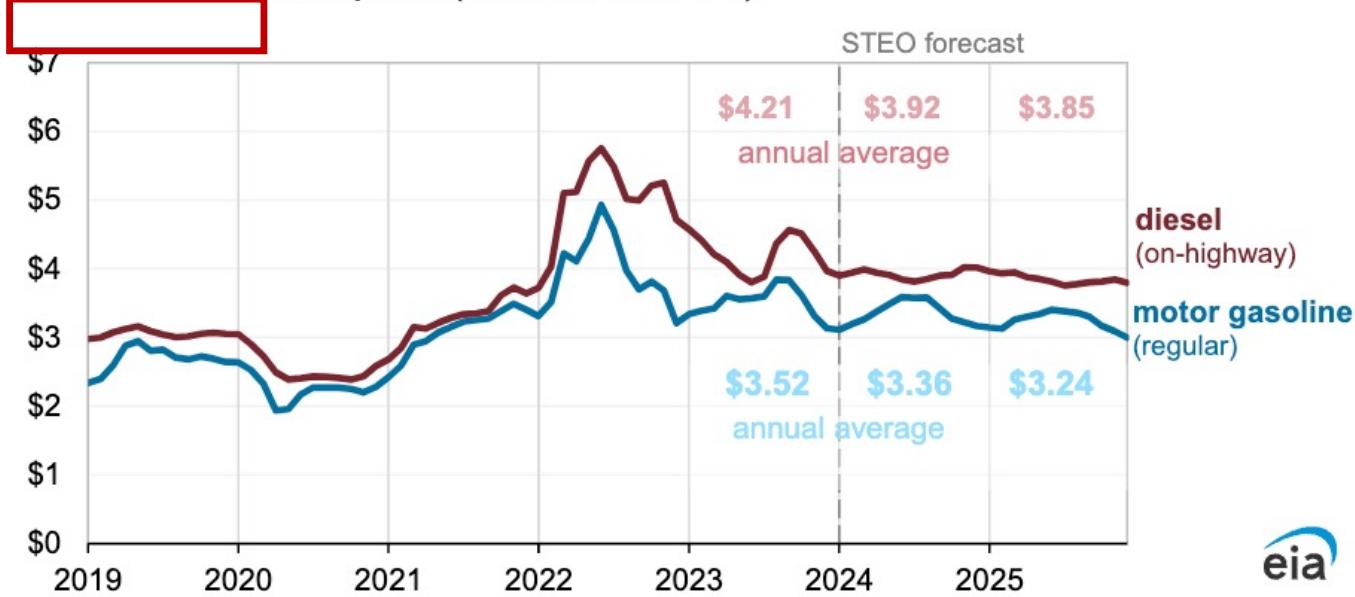
Order of magnitude 数量级

Unit matters



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Monthly U.S. retail fuel prices (Jan 2019–Dec 2025)



Data source: U.S. Energy Information Administration, *Short-Term Energy Outlook* (STEO), January 2024



Is the US gasoline price higher or lower than China's? By how much?

For simplicity, take an exchange rate of 7 (1 USD = 7 RMB)



So we are looking at converting gallon to liter. How much is 1 gallon then?

1 imperial gallon = 4.54609 L

1 US liquid gallon = 3.785411784 L

1 US dry gallon = 4.40488377086 L

Fundamental SI units



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7 fundamental SI quantities and units

Base quantity	Unit	Symbol
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

- The Système International (SI) is the only system of units that is internationally recognized, allowing scientists from all over the planet to use a single standard with which to communicate scientific ideas.
- In the SI, there are seven fundamental quantities, each with its own fundamental unit. (in BP, you need to know the first six quantities and units. How each of these quantities and units is defined is not required though.)

Some derived units with their own names in SI system



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Name	Symbol	Quantity	In SI base units	In other SI units
radian ^[N 1]	rad	plane angle	m/m	1
steradian ^[N 1]	sr	solid angle	m ² /m ²	1
hertz	Hz	frequency	s ⁻¹	
newton	N	force, weight	kg·m·s ⁻²	
pascal	Pa	pressure, stress	kg·m ⁻¹ ·s ⁻²	N/m ²
joule	J	energy, work, heat	kg·m ² ·s ⁻²	N·m = Pa·m ³
watt	W	power, radiant flux	kg·m ² ·s ⁻³	J/s
coulomb	C	electric charge	s·A	
volt	V	electric potential, voltage, emf	kg·m ² ·s ⁻³ ·A ⁻¹	W/A = J/C

Source: [International System of Units - Wikipedia](#)

- Some derived units are given a special name which simply expresses the combination of fundamental units in a more compact form
- There are currently 22 such units in the SI

Scientific Notation - revision



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- In physics, we often need to deal with fairly large or small numbers. E.g.
 - Speed of light in vacuum: $299\,792\,458\text{ m}\cdot\text{s}^{-1}$
 - Distance from the sun to the earth: $150\,000\,000\,000\text{ m}$

To avoid writing so many digits, we can use scientific notation

Scientific Notation



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- Numbers that are too big or too small for decimals are often written in scientific notation

$$a \times 10^b$$

Diagram illustrating the components of scientific notation:

- a is the **Coefficient** ($0 \leq a < 10$).
- 10 is the **Base** (always 10).
- b is the **Exponent** (b is a non-zero integer).

e.g. $153.2 = 1.532 \times 10^2$; $0.00872 = 8.72 \times 10^{-3}$



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Practice on scientific notation

- Express the following in scientific notation (standard form)
 - 19000000
 - 2840
 - 0.00342
 - 0.00004548



Practice on scientific notation

- Express the following in scientific notation (standard form)

- | | |
|--------------|--|
| • 19000000 | 1.9×10^7 (one point nine times ten to the power of seven) |
| • 2840 | 2.84×10^3 |
| • 0.00342 | 3.42×10^{-3} |
| • 0.00004548 | 4.548×10^{-5} |

Metric Multipliers (Prefixes)



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Metric (SI) multipliers

Prefix	Abbreviation	Value
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
hecto	h	10^2
deca	da	10^1
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

- An SI prefix is a name or associated symbol that is written before a unit to indicate the appropriate power of 10
 - 1 000 grams = 1 kilograms (kg)
 - 1 second = 1 000 milliseconds (ms)
 - 1 nanometer (nm) = 1×10^{-9} meters
- A list of metric multipliers is provided in IBDP data booklet that you will get one copy during exams

Orders of magnitude



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- Order of magnitude: A number rounded to the nearest power of 10
- Good for estimating the approximate scope of a quantity. It simply means how big is a number in terms of power(s) of ten

Number	Number in scientific notation	Order of magnitude
230	2.3×10^2	2
890	8.9×10^2	3
0.022	2.2×10^{-2}	-2
0.07	7×10^{-2}	-1

A good tool for checking your answers. If you are asked to work out the mass of an apple, and your answer is $2 \times 10^8 \text{ kg}$, immediately you know something is wrong!